

ADITYA SINGH

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SKILLS

- Technical Skills:** Python, C/C++, SQL, Kafka, Hadoop, PySpark, TensorFlow, Keras, PyTorch, OpenCV, Scikit-learn, Deep Learning and Neural Networks, Data Structures, Algorithms, Data Analysis, Pandas, MongoDB, Docker, Git and GitHub, FastAPI, AWS.
- Soft Skills:** Complex Problem Analysis, Strategic Decision Making, Effective Communication, Leadership.

EXPERIENCE

Software Development Engineer Intern | Amazon

Jan 2026 - June 2026

- Architected an **AI chatbot** on **Bedrock AgentCore** that let **5,000+** users instantly query internal documentation instead of hunting through scattered docs.
- Consolidated fragmented data into a **unified Knowledge Base with RBAC and regex validation**, eliminating duplicate documents and ensuring clean, sanitized data.
- Eliminated critical **AgentCore** startup timeouts by building a **7 stack AWS CDK** infrastructure with custom **Docker** images.
- Built a custom **integration test suite** that fixed inaccurate line coverage tracking for SQS requests, capturing CloudWatch metrics with 100% accuracy.

Machine Learning Intern | Suvidha Foundation

Jul 2023 - Aug 2023

- Fine-tuned** a **Transformer based BERT model**, lifting summarization accuracy by **20%** over the legacy workflow.
- Deployed the upgraded summarization model into production, **cutting data processing time by 30%**.
- Evaluated output quality with **ROUGE** metrics, **driving a 10% increase in ROUGE-L scores**.

PATENTS

Component-Aware Asynchronous Offloading (CAAO)

- Invented a memory orchestration framework that maps OS **virtual memory** concepts onto **deep learning**, using a **4 stage instruction pipeline (Prefetch, Compute, Offload, Update)** across independent hardware streams to **mask PCIe transfer latency** reducing **training time VRAM consumption by upto 62%**.
- Enabled enterprise grade **ViT/GAN** training on consumer GPUs by integrating **zero copy DMA transfers** via **pinned CPU memory**, eliminating bounce buffers and bounding peak VRAM to the largest active component.
- Authoring a research paper on **CAAO's memory orchestration framework** for submission to MLSys.

PROJECTS

LLM Powered DBMS | GitHub

- Spearheaded a **full-stack LLM powered DBMS** that translates natural language into optimized SQL, freeing nontechnical users from writing SQL by hand, **decreasing on-boarding time by 21%**.
- Engineered a scalable **RESTful** backend in **FastAPI** and **SQLite**, hand tuning low level table schema and indexing strategies for sub-second responses even on complex, multi table queries.
- Delivered full **CRUD** support (create, insert, query, update, delete) purely through natural language, including schema generation, **hitting 89.6% correct SQL translation across 500 varied test commands**.

Vision Transformer Compression for Edge Inference | GitHub

- Compressed a **DeiT-based ViT** by combining structured **attention-head/MLP pruning, knowledge distillation**, and **INT8 dynamic quantization**, cutting model size **79.2%** (21.2MB → 4.4MB) while retaining **91.9%** of baseline top-1 accuracy (80.0% → 73.5%).
- Designed a weight-magnitude-based structured pruning method targeting attention heads and MLP hidden units directly, **cutting parameter count 29.8%** (5.54M → 3.89M) across all 12 transformer blocks without disrupting residual connections.
- Recovered pruning-induced accuracy loss through **knowledge distillation** from a **DeiT-Small teacher**, improving accuracy by 8.2 points over pruning alone (66.1% → 74.3%).
- Benchmarked CPU inference latency and accuracy across six compression configurations (fp32/int8 × baseline/pruned/distilled), producing a full accuracy/latency/size Pareto analysis for edge-deployment decisions.

EDUCATION

Indian Institute of Information Technology, Allahabad

M.Tech in Information Technology (Machine Learning) - 9 CGPA

Prayagraj, India

Aug 2024 - July 2026

University Institute of Technology, RGPV

B.Tech in Computer Science and Engineering

Bhopal, India

Sep 2020 - May 2024